Original Article

Maternal health risk analysis, automated pregnancy risk level identification and status monitoring system based on multivariable clinical data

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Background: Ensuring high-quality care and regular monitoring of maternal health are essential during pregnancy to promptly detect potential complications. Unfortunately, the lack of specialist obstetricians and necessary expertise poses challenges in providing frequent follow-up and monitoring for pregnant women in low-resource areas. This research aimed to address these issues by designing and developing a pregnancy risk level and status monitoring system using a machine learning approach.

Methods: A system for classifying pregnancy risk levels and monitoring status was developed using data from pregnant women, including age, systolic and diastolic blood pressure, blood sugar levels, body temperature, and heart rate. Six machine learning models were trained to classify risk levels. Additionally, a web-based user interface was developed to enable physicians to identify risk levels and track the trends of each clinical parameter throughout the pregnancy period.

Results: Based on our experimental analysis, we found a strong correlation between high pregnancy risk and elevated systolic and diastolic blood pressures, higher blood sugar levels, and older ages compared to other clinical parameters. Out of the six machine learning models trained and validated, the random forest classifier demonstrated superior performance with a 91% classification accuracy and was chosen for risk level classification.

Conclusions: Our results highlight the accuracy and effectiveness of the machine learning model in predicting pregnancy risk levels using clinical data. The proposed risk level and status monitoring systems can serve as valuable decision support tools for managing maternal complications, particularly in resource-limited settings.

Keywords: Maternal complications; machine learning model; pregnancy; risk level; low resource settings

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Introduction

Approximately 800 women die every day around the world due to pregnancy and child birth complications that can be prevented (1). Among these, 99% of the estimated global maternal mortality occurs in low resources setting, and half of this is in Sib-Saharan Africa (2-4). This is mainly due to many factors including inadequate healthcare setup, shortage of skilled health workers, poverty, distance to facilities, lack of information, poor quality service, cultural beliefs and practices, and various patient-related factors like the acceptability and affordability of maternal health services (1,5,6).

Primary complications during pregnancy and childbirth leading to maternal mortality and morbidity encompass severe bleeding, infections, high blood pressure, delivery-related complications, unsafe abortion, cardiovascular issues, diabetes, anemia, as well as depression and anxiety (7). The causes of maternal mortality in Sub-Saharan countries are post-partum hemorrhage, hypersensitive pregnancy disorder (e.g., pre-eclampsia/eclampsia), non-obstetric complications, and pregnancy-related infections (e.g., puerperal sepsis) (6). Pregnant women are also at a high risk of infectious disease such as coronavirus disease 2019 (COVID-19) and related complications (8,9). The disparity in maternal mortality reduction between developing and developed countries may indicate the low efficiency of evidence-based interventions in developing countries to detect and manage risk factors of maternal complications.

Several factors can contribute to high-risk pregnancy that leads to perinatal morbidity and mortality. Pregnancy risks can be high or low. High-risk pregnancies can cause a one in four chance of developing complications, while low-risk pregnancies have nearly a one in ten chance of developing complications (10). The mothers' age, high blood pressure, obesity, diabetes, epilepsy, thyroid disease, and heart diseases are among the high-risk pregnancy complications (11-13). Early detection of pregnancy complications through regular quality examinations especially for high-risk pregnancies before, during and after childbirth can reduce maternal morbidity and mortality

Risk assessment before and during pregnancy allows women and care givers to early predict the most likely adverse health events and make informed decisions to prevent it. It constitutes a crucial element of antenatal care (ANC) and is the central focus of maternal and childcare programs, holding the potential to enhance both maternal and perinatal outcomes (16-19). Various risk factor-based scoring or assessment systems are employed throughout the antepartum, intrapartum, and neonatal periods to stratify risks. Antenatal risk scores are numerical indices that reflect the collective risk associated with maternal characteristics. The first antenatal risk score was developed by Goodwin *et al.* (20) in 1969 and validated later by Burstyn (21) which was used to evaluate more than 900 pregnancies. The scores were classified into low, moderate, and high-risk categories for pregnancies. These risk scores play a crucial role in identifying pregnant women at high risk, thereby helping prevent maternal and fetal complications.

Obstetricians/physicians assess and categorize clinical data from clinical and diagnostic investigations to produce risk scores based on their knowledge and expertise. However, the existing approach is very subjective, necessitates interpretation abilities, and is prone to error which may lead to misdiagnosis. Furthermore, the manual assessment method could be tedious and time-consuming. Clinical decision support systems based on machine learning have evolved as a critical component of technology-assisted medical care, assisting numerous clinical procedures through the utilization of experiential knowledge. Clinical

Highlight box

Key findings

- The study found a strong correlation between high pregnancy risk and elevated systolic and diastolic blood pressures, higher blood sugar levels, and older ages.
- Among six machine learning models trained for pregnancy risk classification, the random forest classifier performed best, achieving a classification accuracy of 91%.
- A web-based system was developed to allow physicians to monitor pregnancy risk levels and clinical parameter trends.

What is known and what is new?

- Monitoring pregnancy health is crucial for early detection of complications, but low-resource settings face challenges such as the lack of specialist obstetricians and regular follow-up opportunities.
- This study introduces a machine learning-based system that can accurately classify pregnancy risk levels and monitor maternal health using a web-based interface, providing an efficient solution for managing complications in low-resource areas.

What is the implication, and what should change now?

- The machine learning-based system offers a reliable and accessible tool for healthcare providers in resource-limited settings to monitor pregnant women, improving early detection and management of potential complications.
- Further scaling and implementation of this system in low-resource healthcare settings is necessary to improve maternal care.

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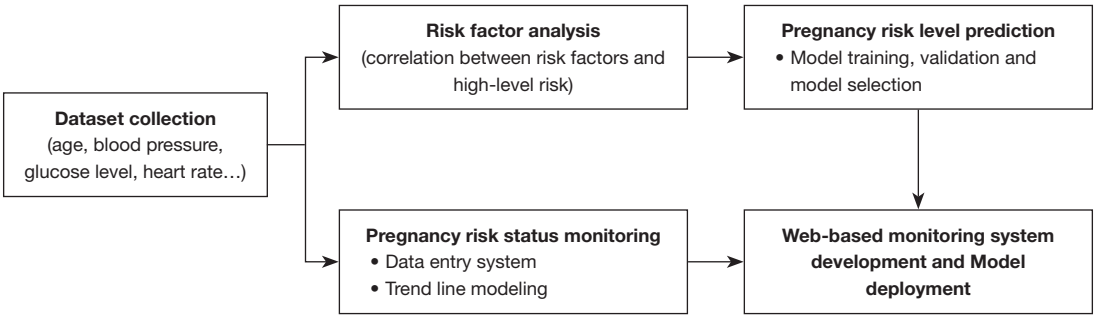

Figure 1 General method framework of the proposed system.

decision support system (CDSS) aims to enhance healthcare service by improving the decision-making process of healthcare professionals by supplementing medical choices with targeted clinical knowledge, patient data, and other health data (22,23).

The current work presents a development and validation of machine learning CDSS for pregnancy risk level assessment, pregnancy risk status monitoring system with the aim of allowing for improved pregnancy management and, as a result, better outcomes for both the mother and the fetus. We present this article in accordance with the TRIPOD reporting checklist (available at https://ht.amegroups.com/article/view/10.21037/ht-23-10/rc).

Methods

The study was conducted in accordance with the Declaration of Helsinki (as revised in 2013). This research did not involve humans, animals, or other subjects. According to Jimma University's Institutional Review Board (IRB), no formal ethics approval was required in this particular case. Because of the retrospective nature of the research, the requirement for informed consent was waived.

The general framework of the proposed system is demonstrated in *Figure 1*. The proposed system includes a pregnancy risk level classification and status monitoring. The pregnancy risk predictor uses patient and clinical data to identify the risk level. The pregnancy risk status monitoring also uses the same attribute information and shows the trend line of each parameter from the start to current pregnancy period. Both of these systems are integrated and deployed into a web-based user interface.

Data

To deploy the proposed pregnancy risk predictor and monitoring system, a dataset comprising 1,014 observations was utilized. Each observation contained distinct attribute information, including age, systolic blood pressure, diastolic blood pressure, blood glucose level, body temperature, and heart rate. The data was gathered from a publicly available database (24) that was collected from various health facilities. *Table 1* displays sample data from 10 pregnant women with annotated risk levels. The data consists of age (years at the time of pregnancy), systolic BP (upper blood pressure measurement in mmHg), diastolic BP (lower blood pressure measurement in mmHg), BS (blood glucose concentration in mmol/L), body temp (body temperature in Celsius), heart rate (resting heart rate in beats per minute), and risk level (categorized risk intensity during pregnancy).

Pregnancy risk factor analysis

Various risk factors play a role in pregnancy complications. This study examines major risk factors, including age, blood pressure (both systolic and diastolic), blood sugar level, body temperature, and heart rate. The analysis focuses on their contributions to high, medium, and low pregnancy risk levels. Box and whisker plots, a standardized method for visualizing data distribution using the five-number summary (minimum, first quartile, median, third quartile, and maximum), were used to represent the distribution of each risk level in relation to the collected data for each risk factor. In addition, to analyze the correlation of each risk factor with each other and with the high-risk level, a correlation matrix was used. Correlation matrix is a table

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Table 1 Sample data collected from 10 pregnant women: 4 labeled high risk, 3 medium risk and 3 low risks

No.	Age (years)	Systolic BP (mmHg)	Diastolic BP (mmHg)	BS (mmol/L)	Body temp (℃)	Heart rate (bpm)	Risk level
1	25	130	80	15	36.6	86	High risk
2	35	140	90	13	36.6	70	High risk
3	35	120	60	6.1	36.6	76	Low risk
4	23	130	70	7.01	36.6	78	Mid risk
5	35	85	60	11	38.8	86	High risk
6	32	120	90	6.9	36.6	70	Mid risk
7	42	130	80	18	36.6	70	High risk
8	23	90	60	7.01	36.6	76	Low risk
9	19	120	80	7	36.6	70	Mid risk
10	25	110	89	7.01	36.6	77	Low risk

BP, blood pressure; BS, blood glucose.

that shows the correlation coeffects between variables.

Training and testing risk level prediction models

In this study, five machine learning models were trained and tested including K-nearest neighbor (KNN), multiclass logistic regression (MLR), support vector machine (SVM), extreme gradient boosting (XGBoost), and random forest—alongside a deep learning model, the multi-layer perceptron neural network (MLP), for classifying pregnancy risk levels. To evaluate the performance of each model and identify the best one, the same dataset was used for both training and testing across all classifiers.

KNN is a non-parametric, supervised algorithm that can be applied to both classification and regression tasks. For classification, it assigns a class to an object based on the majority vote of its nearest neighbors. Multiclass or multinomial regression extends logistic regression to handle problems with multiple classes by altering the loss function to cross-entropy and predicting a probability distribution across several outcomes. XGBoost, on the other hand, is an ensemble learning algorithm that uses decision trees within a gradient boosting framework. Known for its efficiency, flexibility, and scalability, it is a highly optimized distributed gradient boosting library. XGBoost applies machine learning algorithms within the Gradient Boosting framework, offering parallel tree boosting to improve efficiency, flexibility, and portability (25). For this paper, the XGBoost model was trained using a learning rate set at 0.001, an L1 regularization value of 5, an L2 regularization value of 2, 2,000 estimators or runs (model learning iterations), and a maximum depth of 1,000. Random forest, on the other hand, is a supervised learning algorithm employed for both classification and regression tasks (26). The random forest algorithm was trained with 2,000 decision trees (estimators), a depth of 1,000, and default values for other parameters. On the other hand, the multilayer perceptron represents a class of neural networks and serves as the fundamental form of a deep neural network, comprising a series of fully connected layers. These models are employed to address the typically high computing power demands associated with state-of-the-art deep learning architectures.

Data augmentation was conducted before initiating model training to mitigate class imbalance. Classifying imbalanced data may introduce bias in favor of the majority class, leading to suboptimal performance on the minority class. The Synthetic Minority Oversampling Technique (SMOTE) was employed on the dataset to address class imbalance. SMOTE is an oversampling method that involves generating "synthetic" examples for the minority class. This is achieved by creating synthetic examples along the line segments connecting any or all of the KNNs of each minority class sample (27). This approach aids in mitigating the overfitting issue associated with random oversampling. Following class balancing through SMOTE, the data was divided into a train/test split of 70/30, with 70% allocated for training and 30% for testing.

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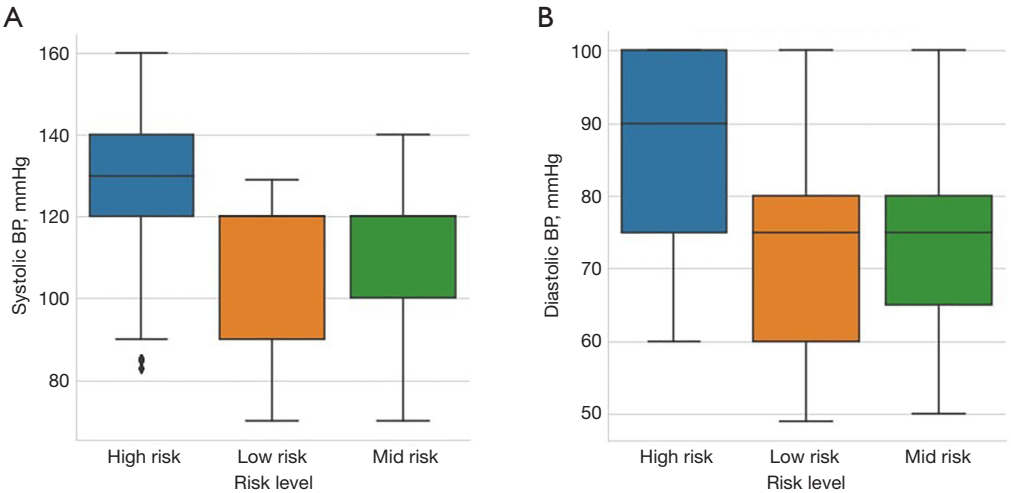

Figure 2 Distribution of risk level based on (A) systolic and (B) diastolic BP. BP, blood pressure.

Performance evaluation metrics

The performance of the trained models was assessed and compared using accuracy, precision (also referred to as positive predictive value), sensitivity (also known as recall), F1 score (the harmonic mean of sensitivity and precision), and receiver operating characteristic (ROC)-area under the curve (AUC) plot. These evaluation metrics were calculated from the model's confusion matrix, which summarizes classification outcomes using TP (true positive), TN (true negative), FP (false positive), and FN (false negative) values, as shown in Eqs. [1-6].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad [1]$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad [2]$$

$$\text{Precision} = \frac{TN}{TN + FP} \quad [3]$$

$$\text{Specificity} = \frac{TP}{TN + FP} \quad [4]$$

$$\text{F1 score} = \frac{2 * \text{Recall} * \text{precision}}{\text{Recall} + \text{precision}} \quad [5]$$

Results

Pregnancy risk factor analysis

Figures 2,3 illustrate the distribution of pregnancy risk levels based on the risk factors [blood pressure (mmHg), heart rate (beats per minute), body temperature (℃), blood sugar level (mmol/L) and age] from the collected data. As indicated, high-risk factors are more visibly distributed at high systolic and diastolic blood pressures, high blood sugar levels and older ages.

The correlation between the pair of risk factors (features), and each risk factor with the high-risk level is demonstrated in *Figure 4*. Each cell in the correlation table shows the correlation between two variables (risk factors). As demonstrated, blood sugar level has the highest correlation with the high-risk level followed by diastolic and systolic blood pressure, while temperature has the lowest correlation with the high-risk level.

Results of risk level classification models

Summary of the trained classification models accuracy, precision, sensitivity, and F1 score is demonstrated in *Table 2*. The random forest classifier showed the highest performance

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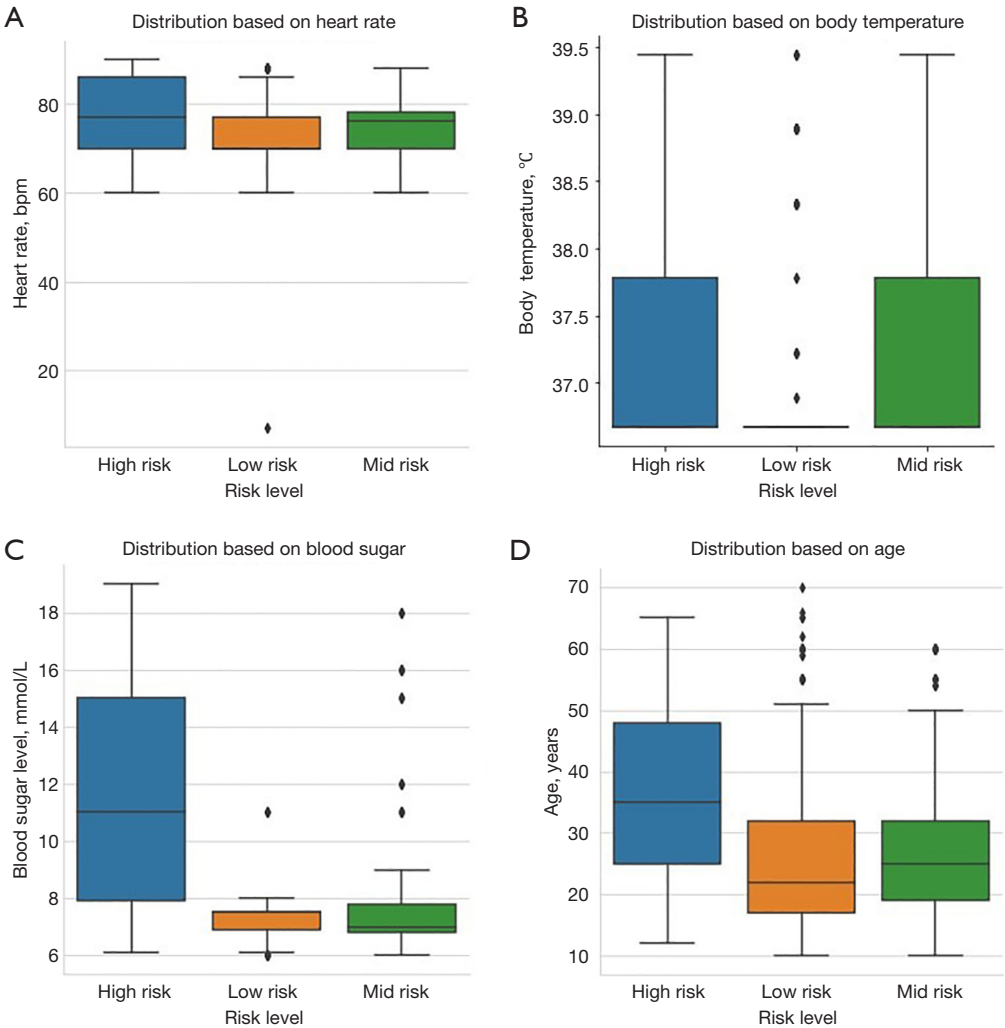

Figure 3 Distribution of risk level based on (A) heart rate, (B) body temperature, (C) blood sugar level, and (D) age.

in all evaluation metrics with 91% accuracy, 91.6% precision, 90.6% sensitivity and 91% F1 score in classifying the test data into high-risk, medium-risk and low-risk followed by KNN (88% accuracy). Hence, the random forest classifier was selected for implementation. The ROC curves of the random forest classifier evaluated on test data and the confusion matrix showing true positive and false negative values are demonstrated in *Figures 5,6*, respectively.

User interface

For ease of use of the developed risk classification and pregnancy monitoring system, a user interface was developed using an open-source browser-based Python framework called '*Streamlit*'. The user interface has two main pages: pregnancy risk level prediction page (*Figure 7*) and pregnancy risk status monitoring (*Figure 8*). The risk level prediction system accepts patient's age, systolic and diastolic blood pressure, blood sugar level, body temperature and heart rate, and presents the predicted risk level along with the percentage of prediction. Users can then record the clinical parameters to an existing patient or a new patient as a history file. Using the 'pregnancy risk status monitoring page', users can select a patient name from the drop-down list and observe the trend of each clinical parameter graphically from initial record date to the present date. This allows physicians or caregivers to track the trend of each parameter, monitor which parameters went beyond the normal range, throughout the pregnancy period and to monitor and make an informed decision.

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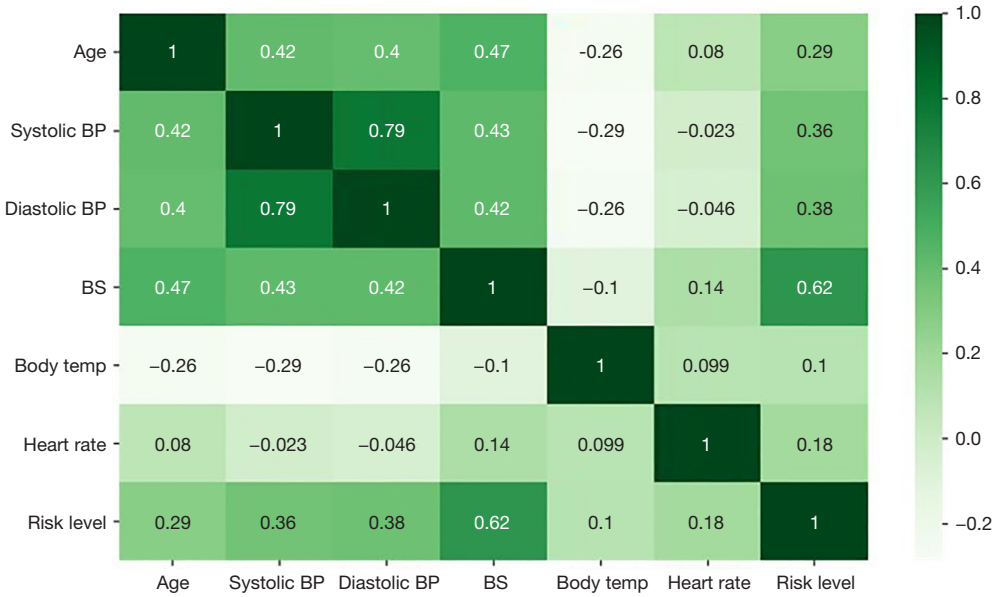

Figure 4 Correlation matrix between features/risk factors and with high-risk level. BP, blood pressure; BS, blood glucose.

Table 2 Summary of performances of classification models

Models	Precision (%)	Sensitivity (%)	F1 score (%)	Accuracy (%)
KNN	88	87.3	87.3	88
MLR	65	65	63.3	64
SVM	64	61.6	62	61
MLP	73	63	57	63
XGBoost	83	83	83	83
Random forest	91.6	90.6	91	91

KNN, K-nearest neighbor; MLR, multiclass logistic regression; SVM, support vector machine; MLP, multi-layer perceptron neural network; XGBoost, extreme gradient boosting.

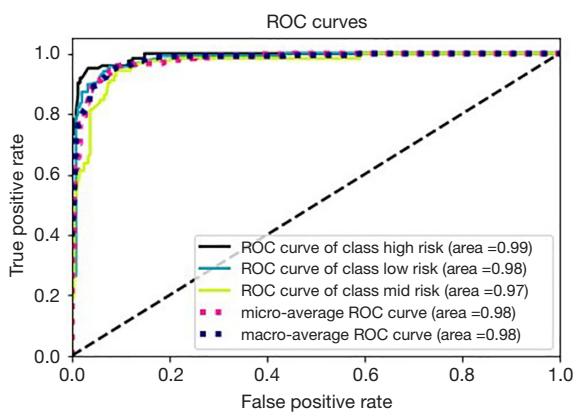

Figure 5 ROC curves of the random forest classifier on test data. ROC, receiver operating characteristic.

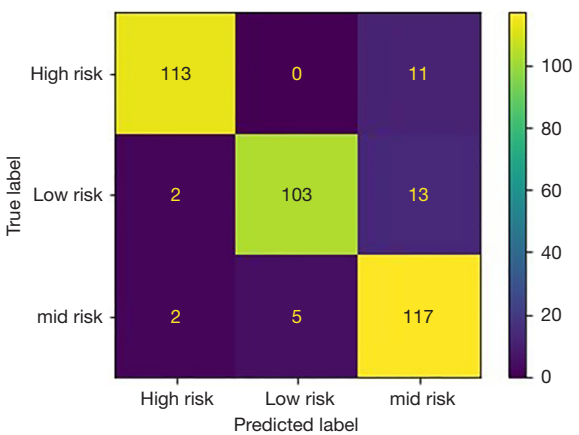

Figure 6 Confusion matrix demonstrating the number of true classes and predicted classes by the random forest classifier.

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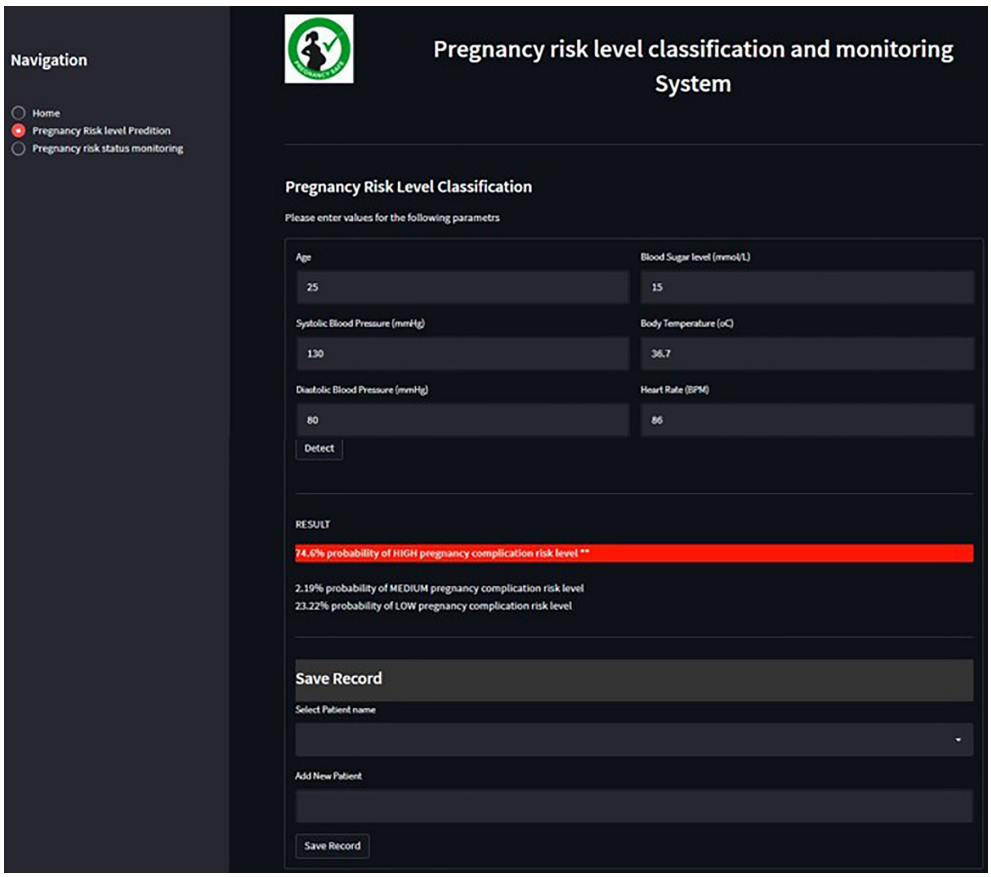

Figure 7 Snapshot of the pregnancy level prediction system user interface.

Discussion

Every year, many pregnant women in underdeveloped countries, notably in Sub-Saharan Africa, die from pregnancy-related morbidities (28). The common leading causes of pregnancy-related mortality and morbidity are the pregnancy complications including blood pressure, gestational diabetes, preeclampsia (29), anemia, infections etc. It might sometimes be difficult to tell the difference between normal pregnancy symptoms and symptoms of problems. Early detection and proper management of these pregnancy-related risk problems is critical for providing effective care to the pregnant women. For effective results, maternal risk assessment and the subsequent management should be guided by evidence-based practice. However, with limited resources and experts, which are the cases of least developed countries, early pregnancy risk assessment and the resulting management is not up to the level of the burden and sensitivity of the issue.

The integration of digital technology into healthcare systems has opened the door for the potential enhancement of disease diagnosis and management through AI powered clinical decision support systems. These systems have the capability to improve accuracy, speed, and efficiency. One fundamental advantage of AI or machine learning systems is their objectivity, as they rely on real-world data and results, identifying crucial variables for physicians. AI stands poised to revolutionize women's healthcare by enhancing diagnosis, alleviating physician workload, reducing healthcare costs, and offering benchmark analysis for tests with notable interpretation discrepancies among specialists. This holds particular significance for developing countries where expertise and resources are often limited.

This paper introduces a machine learning-driven system for classifying pregnancy risk levels and monitoring status using vital clinical parameters. The analysis and training of machine learning models for pregnancy risk level classification, as well as the development of the pregnancy risk status monitoring system, were conducted using data

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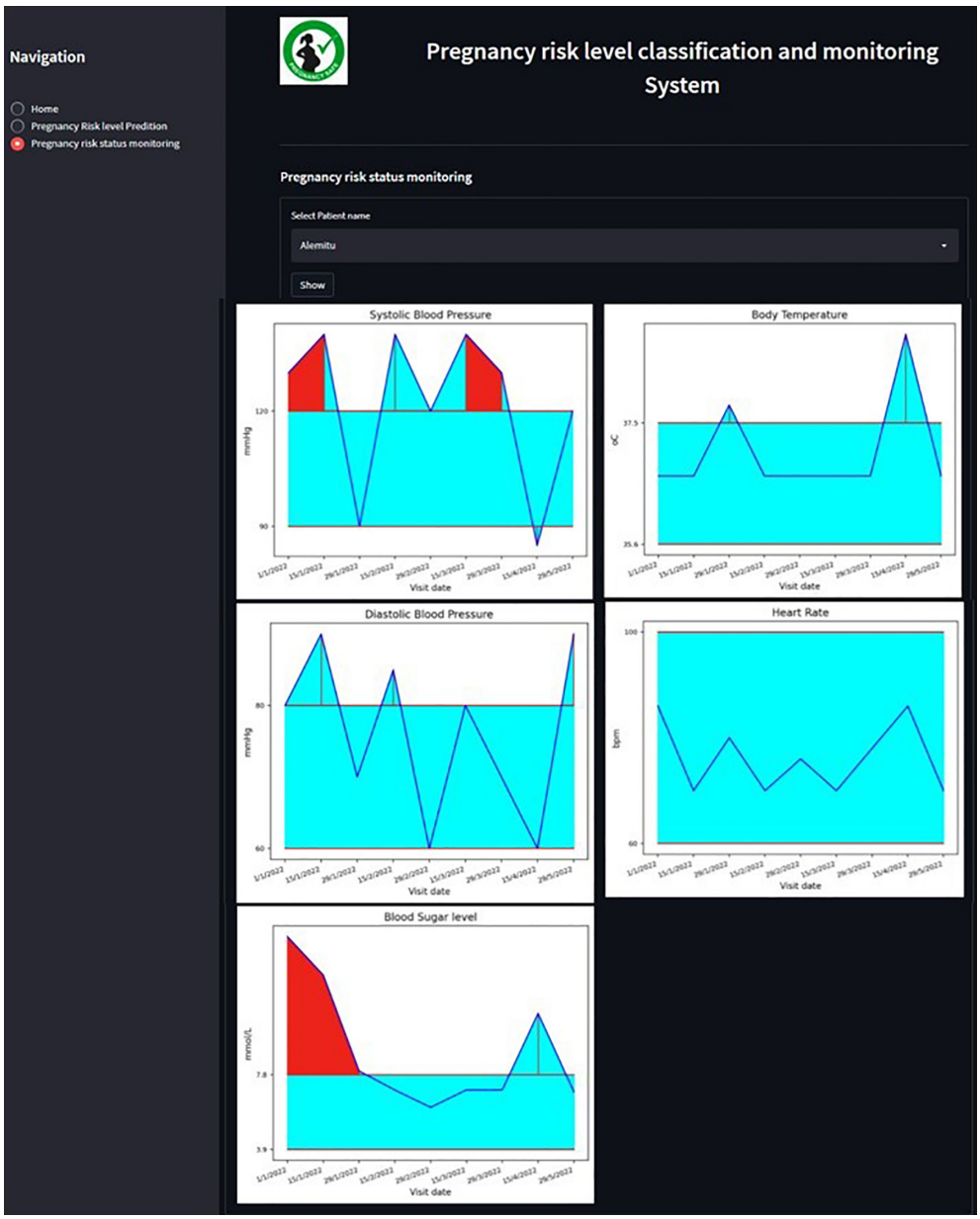

Figure 8 Snapshot of the pregnancy risk status monitoring system user interface.

comprising age, systolic and diastolic blood pressure, blood sugar level, body temperature, and heart rate.

The analysis from our data indicates that high pregnancy risk is highly associated with higher systolic blood pressure, diastolic blood pressure, blood sugar level and older (*Figures 2,3*). This is evident that high blood pressure can cause a variety of health concerns, including cardiovascular disease, and it can disrupt the placenta's development, limiting the baby's nutrient and oxygen supply (30-33). The chance of having high blood pressure (hypertension) also increases with age and presence of diabetes mellitus (34-37). On the other hand, high blood glucose levels during pregnancy can potentially lead to premature labor, excessive fetal weight, or newborn respiratory issues after birth (38,39). This is also demonstrated in correlation plot of *Figure 4*, with blood sugar level having the highest correlation with the high-risk level followed by diastolic and systolic blood pressure, while temperature has the lowest

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correlation compared to other clinical parameters analyzed from the data used in this study. This indicates that the most important features for discriminating pregnancy risk levels, according to our data, are blood sugar level, systolic blood pressure and diastolic blood pressure.

The proposed machine learning based pregnancy risk level classification and monitoring system can be used as a decision support system to predict pregnancy risks early and manage them properly. Using the user interface, the physicians/obstetricians can feed vital clinical parameters to the classification system and automatically determine the risk level of pregnancy. The system will return the percentage of prediction probability for each class to allow the user to make an informed decision based on the results. The results along with the clinical records can also be saved for later reference and as an input to the pregnancy risk status monitoring system. Using the risk status monitoring system, user can regularly follow up the trends of each clinical parameter graphically throughout pregnancy period and monitor which parameters went below or above the normal ranges. This is significant to early detect health problems that could arise at any period of pregnancy, provide evidence-based treatment and increase the chance for normal pregnancy and a healthy baby's birth. We acknowledge the need for a model trained on extensive and diverse datasets before it can be applied in clinical settings.

In conclusion, the proposed system can serve as a decision support tool, aiding healthcare professionals in the early prediction of pregnancy risk levels and improving pregnancy management. Moreover, it can be utilized to maximize the distribution of information to physicians/obstetricians at a point in the clinical workflow where decision-making is most effective. We acknowledge that further research by including a greater number of clinical parameters on a preferably much large number of data is required for optimal use of the proposed system.

Conclusions

This paper introduces a machine learning-based system for classifying pregnancy risk levels and monitoring status using clinical parameters. The study analyzed the correlation between clinical parameters and high pregnancy risk, identifying blood sugar level, systolic blood pressure, and diastolic blood pressure as crucial features for risk discrimination. Six machine learning models were trained and validated on a dataset of 1,014 entries, consisting of age, blood pressure, blood glucose level, body temperature, and heart rate for pregnancy risk prediction. The random forest model, which achieved a classification accuracy of 91%, was chosen for risk level classification. A user-friendly web-based interface was developed for both the pregnancy risk level classification and status monitoring systems, serving as a decision support tool for healthcare professionals in early pregnancy risk prediction and management.

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Footnote

Reporting Checklist: The authors have completed the TRIPOD reporting checklist. Available at https://ht.amegroups. com/article/view/10.21037/ht-23-10/rc

Data Sharing Statement: Available at https://ht.amegroups.com/article/view/10.21037/ht-23-10/dss

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki (as revised in 2013). This research did not involve humans, animals, or other subjects. According to Jimma University's Institutional Review Board (IRB), no formal ethics approval was required in this particular case. Because of the retrospective nature of the research, the requirement for informed consent was waived.

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